

Amira Mouamine

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EDUCATION

Université de Montréal

Montreal, QC

M.Sc. in Computer Science, Specialization in Artificial Intelligence (thesis-based)

Sep. 2026 – Expected May 2028

Université Laval

Quebec, QC

B.Eng. in Computer Engineering

Aug. 2021 – May 2026

- Relevant Coursework: Computer Graphics, Game Development Programming, Machine Learning, Real-Time Embedded Systems, Database Models and Languages

TECHNICAL SKILLS

Programming Languages: C++, C#, C, Python, JavaScript/TypeScript, Rust, Java, GLSL, SQL, VHDL, MATLAB

Frameworks & Libraries: OpenGL, openFrameworks, OpenCV, React, Node.js, scikit-learn, PyTorch, TensorFlow

Tools: Git, Azure DevOps, Docker, Azure Blob Storage, Unity, Unreal Engine, Blender, MySQL, Xilinx Vivado

EXPERIENCE

Bentley Systems

May 2025 – Aug. 2025

Full Stack Developer Intern

- Delivered a production **manual measurement feature**, unlocking **long-requested flexibility** for civil engineers by enabling measurements to persist across sessions.
- Owned and implemented the **end-to-end measurement lifecycle** across multiple views (create, auto-save, rename, duplicate, delete, recover), **reducing data loss and rework**.
- Integrated secure cloud storage with **Azure Blob Storage and SAS tokens**, while testing and debugging the feature in an Agile environment.

PROJECTS

Movie Recommendation System | *Python, scikit-learn*

- Built a hybrid recommendation system combining clustering, semantic similarity and supervised rating prediction.
- Processed movie descriptions and genres to generate recommendations based on user preferences.

Graphics Editor | *C++, GLSL, OpenGL, openFrameworks*

- Built support for interactive **3D** scenes (camera controls, lighting, materials, shaders) alongside **2D** drawing tools.
- Enabled image and **3D model import**, real-time rendering and render export for interactive visualization.

Cat & Rats Game | *C#, Unity, Aseprite*

- Built a **two-player co-op game** with custom controls, procedural level generation and a multi-scene game flow.
- Implemented an **AI enemy** using A* pathfinding and state-based reactive behavior.
- Developed **avatar customization** and **audiovisual feedback** (event-driven animations, VFX, SFX, music).

Motion Logger | *Rust, OpenCV, Warp*

- Implemented **real-time motion detection** using grayscale frame differencing with **OpenCV**.
- Built a modular system (camera, server, logger) exposing motion data via a **Warp HTTP server**.

ACTIVITIES & LEADERSHIP

Artificial Intelligence Club of Université Laval

Sep. 2025 – May 2026

Member

- Contributed to an EMG signal processing and ML-based **gesture recognition** project that received the **Coup de Cœur Award**, enabling **real-time hand-controlled** driving from simulation testing to **F1TENTH** deployment.

Game Development Club of Université Laval

Sep. 2024 – May 2026

Co-Leader

- Organized game jams, industry talks and technical workshops for a community of approximately **30 members**.
- Designed and led a hands-on **Unreal Engine workshop**, guiding participants in implementing movement, collisions, scoring and UI using Blueprints and UMG.
- Represented the Faculty of Science & Engineering at **Eurêka! Festival** and at the **Girls & Science** event.